Channel 1

File: channel1.kicad_sch

Channel 2

File: channel2.kicad_sch

CSI

File: csi.kicad_sch

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antmicro

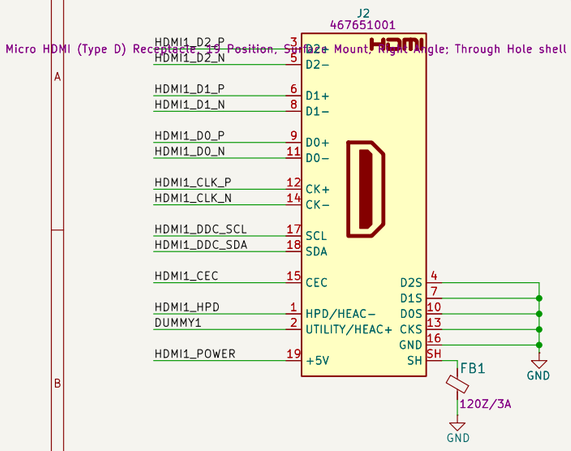
Logo N1 antmicro_logo Logo N2 oshw_logo

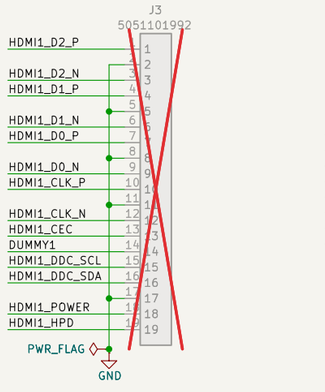
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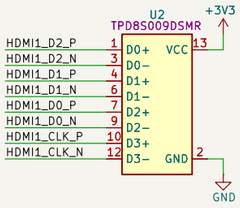
Title: HDMI-MIPI Bridge

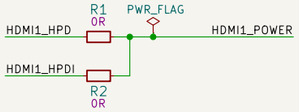
Size: A3	Date: 2024-04-24	**Rev: 1.3.2**
KiCad E.D.A. 10.0.7		Id: 1/4

Connectors

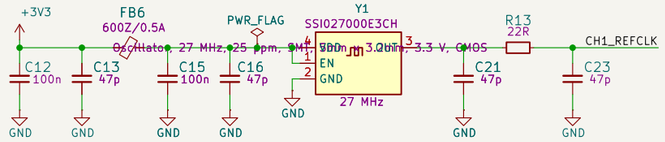




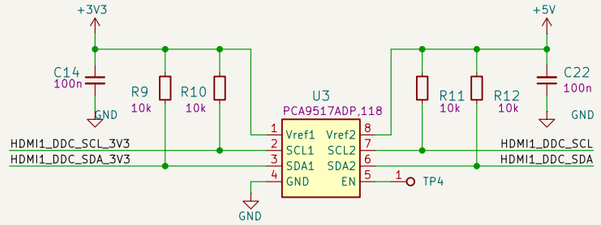




27MHz Oscillator

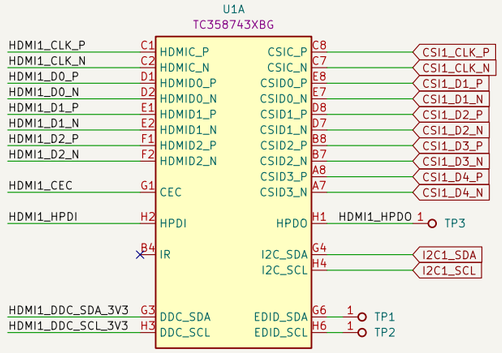


I2C Level Shifter

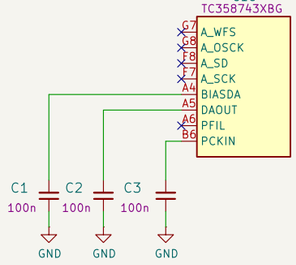


Converter IC

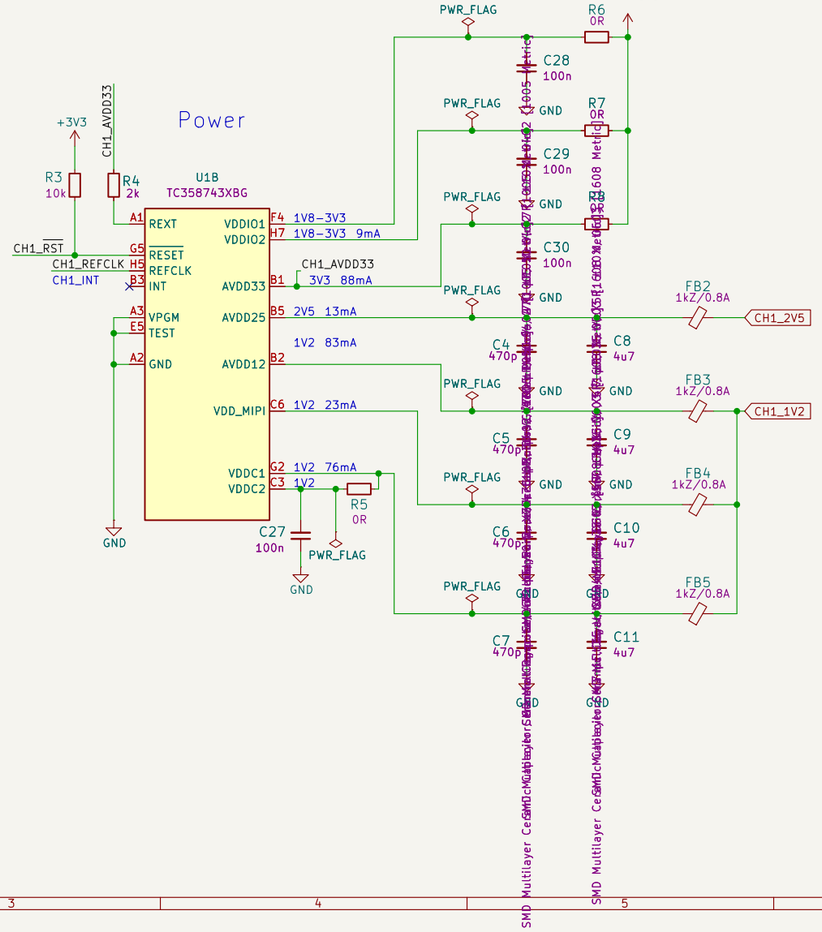
Video Interfaces



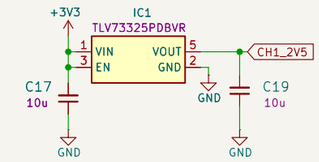
Misc

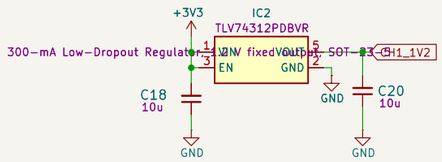


Power



Voltage Regulators





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Sheet: /Channel 1/		
File: channel1.kicad_sch		
Title: HDMI−MIPI Bridge		
Size: A3	Date: 2023−07−20	**Rev: 1.3.1**
KiCad E.D.A. 10.0.7		Id: 2/4

Connectors

J4
467651001
Micro HDMI (Type D) Receptacle 19 Position, Surface Mount, Right Angle; Through Hole shell

HDMI2_D2_P 3 D2+
HDMI2_D2_N 5 D2−
HDMI2_D1_P 6 D1+
HDMI2_D1_N 8 D1−
HDMI2_D0_P 9 D0+
HDMI2_D0_N 11 D0−
HDMI2_CLK_P 12 CK+
HDMI2_CLK_N 14 CK−
HDMI2_DDC_SCL 17 SCL
HDMI2_DDC_SDA 18 SDA
HDMI2_CEC 15 CEC
HDMI2_HPD 1 HPD/HEAC−
DUMMY2 2 UTILITY/HEAC+
HDMI2_POWER 19 +5V
D2S 4
D1S 7
D0S 10
CKS 13
GND 16
SH GND
FB7 120Z/3A
GND

J5
5051101992

HDMI2_D2_P
HDMI2_D2_N
HDMI2_D1_P
HDMI2_D1_N
HDMI2_D0_P
HDMI2_D0_N
HDMI2_CLK_P
HDMI2_CLK_N
HDMI2_CEC
DUMMY2
HDMI2_DDC_SCL
HDMI2_DDC_SDA
HDMI2_POWER
HDMI2_HPD
GND

U5
TPD8S009DSMR
+3V3

HDMI2_D2_P 1 D0+ VCC 13
HDMI2_D2_N 3 D0−
HDMI2_D1_P 4 D1+
HDMI2_D1_N 6 D1−
HDMI2_D0_P 7 D2+
HDMI2_D0_N 9 D2−
HDMI2_CLK_P 10 D3+
HDMI2_CLK_N 12 D3− GND 2
GND

R14 0R PWR_FLAG
HDMI2_HPD → HDMI2_POWER
HDMI2_HPDI
R15 0R

27MHz Oscillator

+3V3
FB12 600Z/0.5A
PWR_FLAG
Y2 SSI027000E3CH
Oscillator, 27 MHz, ±25 ppm, SMT, 3.0mm × 3.3 V, CMOS
R26 22R
CH2_REFCLK
C35 100n
C36 47p
C38 100n
C39 47p
27 MHz
C44 47p
C46 47p
GND

I2C Level Shifter

+3V3
+5V
C37 100n
R22 10k
R23 10k
U6
PCA9517ADP
R24 10k
R25 10k
C45 100n
HDMI2_DDC_SCL_3V3
HDMI2_DDC_SDA_3V3
HDMI2_DDC_SCL
HDMI2_DDC_SDA
TP8
GND

Video Interfaces

Converter IC

U4A
TC358743XBG

HDMI2_CLK_P C1 HDMIC_P
HDMI2_CLK_N C2 HDMIC_N
HDMI2_D0_P D1 HDMID0_P
HDMI2_D0_N D2 HDMID0_N
HDMI2_D1_P E1 HDMID1_P
HDMI2_D1_N E2 HDMID1_N
HDMI2_D2_P F1 HDMID2_P
HDMI2_D2_N F2 HDMID2_N
HDMI2_CEC G1 CEC
HDMI2_HPDI H2 HPDI
B4 IR
HDMI2_DDC_SDA_3V3 G3 DDC_SDA
HDMI2_DDC_SCL_3V3 H3 DDC_SCL

CSIC_P C8 → CSI3_CLK_P
CSIC_N C7 → CSI3_CLK_N
CSID0_P E8 → CSI3_D1_P
CSID0_N E7 → CSI3_D1_N
CSID1_P D8 → CSI3_D2_P
CSID1_N D7 → CSI3_D2_N
CSID2_P B8 → CSI3_D3_P
CSID2_N B7 → CSI3_D3_N
CSID3_P A8 → CSI3_D4_P
CSID3_N A7 → CSI3_D4_N
HPDO H1 HDMI2_HPDO 1 TP7
I2C_SDA G4 → I2C3_SDA
I2C_SCL H4 → I2C3_SCL
EDID_SDA G6 1 TP5
EDID_SCL H6 1 TP6

Misc

U4C
TC358743XBG

A_WFS G7
A_OSCK G8
A_SD G9
A_SCK G10
A4 BIASDA
A5 DAOUT
A6 PFIL
B6 PCKIN

C24 100n
C25 100n
C26 100n
GND

Power

+3V3
CH2_AVDD33
R16 10k
R17 2k
CH2_RST
CH2_REFCLK
CH2_INT

U4B
TC358743XBG

A1 REXT
G5 RESET
H5 REFCLK
B3 INT
A3 VPGM
E5 TEST
A2 GND

VDDIO1 F4 1V8−3V3
VDDIO2 H7 1V8−3V3 9mA
AVDD33 B1 CH2_AVDD33 3V3 88mA
AVDD25 B5 2V5 13mA
AVDD12 B2 1V2 83mA
VDD_MIPI C6 1V2 23mA
VDDC1 G2 1V2 76mA
VDDC2 C3 1V2
R18 0R
C31 100n
PWR_FLAG
GND

PWR_FLAG
R19 0R
+3V3
C32 100n
R20 0R
C33 100n
R21 0R
C34 100n
FB8 1kΩ/0.8A
CH2_2V5
C47 470p
C51 4u7
FB9 1kΩ/0.8A
CH2_1V2
C48 470p
C52 4u7
FB10 1kΩ/0.8A
C49 470p
C53 4u7
FB11 1kΩ/0.8A
C50 470p
C54 4u7
GND

Voltage Regulators

2.5V

+3V3
IC3
TLV73325PDBVR
VIN VOUT 5 CH2_2V5
EN GND 2
C40 10u
C42 10u
GND

1.2V

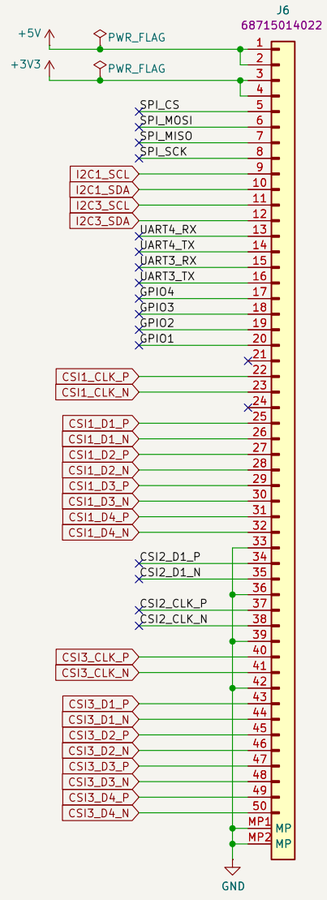
+3V3
IC4
TLV74312PDBVR
300−mA Low−Dropout Regulator, 2% fixed Output, SOT−23
VIN VOUT 5 CH2_1V2
EN GND 2
C41 10u
C43 10u
GND

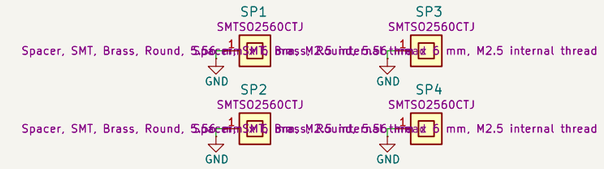
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Sheet: /Channel 2/	
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Title: HDMI−MIPI Bridge	
Size: A3	Date: 2023−07−20
KiCad E.D.A. 10.0.7	**Rev: 1.3.1**
	Id: 3/4

Camera board connector





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Sheet: /CSI/	
File: csi.kicad_sch	
Title: HDMI–MIPI Bridge	
Size: A3	Date: 2023–07–20
KiCad E.D.A. 10.0.7	Id: 4/4

Rev: 1.3.1